

# **Responses to Reviewers**

Manuscript ID: PONE-D-26-01398

“Using a Mixed Opinion Dynamics and Innovation Diffusion Model to Explore the ‘Best Game No One Played’ Phenomenon”

Chung-Yuan Huang and Sheng-Wen Wang

Dear Dr. Malarz,

Thank you for working with our manuscript and your offer for us to revise and resubmit. Thanks also to both reviewers for their constructive evaluations and feedback. A list of the ten most important changes made to our original manuscript would include:

- a revised theoretical framework that explains our decision to integrate bounded confidence opinion dynamics with adoption thresholds, and how the model fits with the Hegselmann-Krause and Deffuant models;
- new controlled experiments aimed at distinguishing the coordination failure channel from the opinion clustering channel;
- finite-size scaling experiments at  $N = 900$ ,  $N = 1,600$  and  $N = 2,500$  to confirm robustness;
- a new discussion section comparing BCAT with alternative frameworks, including information cascades, network externalities, and global games;
- corrected and expanded statistical reporting in Table 3 (algorithm identification, standard errors, and verified data alignment);
- an operational definition for the critical point within the BCAT framework;
- new replacement figures for NetLogo GUI screenshots with author-generated visualizations;
- PLOS LaTeX template conversion with public code and data repositories;
- an expanded Discussion section that adds detail (specific model parameters and simulation figures) for the three motivating cases (the Palm handheld device, the Taiwanese film *The Bold*, *The Corrupt* and *The Beautiful*, and service-oriented architecture [SOA]);
- a comparison of voter models with inertia and SIS/SIR epidemic frameworks to clarify BCAT's unique two-channel structure.

Below please find point-by-point responses to all reviewer comments. Should you have any other concerns or questions, please do not hesitate to contact us.

Sincerely,

Chung-Yuan Huang and Sheng-Wen Wang

# Part I: Journal Requirements

1. Please ensure that your manuscript meets PLOS ONE's style requirements, including those for file naming.

## Response:

We believe this revised manuscript fully complies with PLOS ONE style requirements, including the LaTeX template (plos2025.bst, version 3.7), figure file names (fig1.tif–fig12.tif), multi-panel figures combined with embedded panel labels, TIFF format at 300 DPI, and proper in-text citations.

2. Please note that PLOS One has specific guidelines on code sharing for submissions in which author-generated code underpins the findings in the manuscript. In these cases, we expect all author-generated code to be made available without restrictions upon publication of the work.

## Response:

All author-generated code is publicly available without restrictions under the MIT License at <https://github.com/canslab1/BCAT>, including both the original NetLogo model and a Python 3 reimplementation with pre-configured parameter files for reproducing all reported results. We archived the repository on Zenodo (DOI: 10.5281/zenodo.19216365) for long-term accessibility. A step-by-step protocol for reproducing all simulation results has been published on protocols.io (DOI: 10.17504/protocols.io.261geykydv47/v1). We updated the Data Availability Statement accordingly.

3. Please update your submission to use the PLOS LaTeX template.

## Response:

We prepared the revised manuscript using the official PLOS ONE LaTeX template (plos2025.bst, Version 3.7), including all required formatting elements.

4. Please confirm at this time whether or not your submission contains all raw data required to replicate the results of your study. Authors must share the “minimal data set” for their submission.

## Response:

We confirm that all data required to replicate our results are publicly available. As a computational simulation study, the minimal dataset consists of: (1) complete source code and parameter configuration files (GitHub and Zenodo, as noted above), and (2) raw sensitivity analysis output data underlying Table 3 and Figs 7–9, available in the data/ directory at <https://github.com/canslab1/BCAT>. We updated our Data Availability Statement accordingly.

5. We note that Figures 2, 4, 5, 6, 10 and 11 in your submission contain copyrighted images. We require you to either (1) present written permission from the copyright holder to publish these figures specifically under the CC BY 4.0 license, or (2) remove the figures from your submission.

**Response:**

We have removed the figures in question. In addition, we re-implemented the BCAT model in Python 3 and regenerated all six figures (Figs 2, 4, 5, 6, 10, 11) using our own visualization code. The revised figures present equivalent simulation results under identical parameter configurations. All visual elements are entirely our original work, and no third-party copyrighted materials remain.

## Part II: Responses to Reviewer #1 comments

### R1-1. What Makes the Two Models Combine Organically?

*The paper motivates the BCAT model as a combination of opinion dynamics and adoption thresholds, but the theoretical rationale for why these two components interact in the specific way they do could be articulated more sharply. In particular, the current presentation treats the two components somewhat modularly: agents first update opinions via bounded confidence, then check an adoption threshold. The reader is left to infer the interaction structure from the pseudo-code rather than from a conceptual discussion.*

*It would strengthen the paper to discuss explicitly: (a) what features of bounded confidence opinion dynamics – as opposed to, say, DeGroot-style averaging or voter model dynamics – make it a natural complement to threshold-based adoption, and (b) what structural properties of the combined model are not present in either component alone.*

#### **Response:**

Regarding point (a), in this revised manuscript we discuss why bounded confidence, rather than DeGroot-style averaging or voter model dynamics, represents a natural complement to threshold-based adoption. DeGroot-style averaging models assume that all agents are affected by influences from all neighbors, resulting in guaranteed convergence toward consensus under mild connectivity conditions (Golub & Jackson, 2010, 2012; Cheng et al., 2026). This property makes the DeGroot framework ill-suited for studying the “best game no one played” phenomenon because it cannot generate opinion fragmentation – a key mechanism through which positive average sentiment can coexist with insufficient local support for adoption. Voter model dynamics are discrete and lacking of memory (with each agent randomly copying a neighbor’s binary state), and therefore cannot capture the gradual, continuous nature of attitude evolution that characterizes real consumer sentiment formation. By restricting opinion exchanges to agent pairs whose attitudes fall within a tolerance threshold, bounded confidence naturally generates both consensus (when the confidence bound is large) and fragmentation into multiple clusters (when small). This dual capacity makes bounded confidence suitable for the adoption threshold mechanism, because fragmentation creates the possibility that average attitudes are positive yet local neighborhoods lack sufficient positive-attitude agents to trigger adoption cascades.

Regarding point (b), in this revised manuscript we identify emergent properties of the combined model that are not produced by either component model alone. Most important is the bidirectional feedback loop between adoption status and opinion exchange. Opinion exchange rules in BCAT depend on the adoption statuses of both interacting agents: when one agent has adopted and the other has not, the exchange follows asymmetric persuasion logic (the adopting agent exerts stronger influence). When

both adopt, attitudes converge toward the more positive pole, and when neither adopt, standard bidirectional convergence applies. The adoption decision (a downstream consequence of opinion formation) thus offers feedback to the opinion dynamics process, creating a coupled dynamical system whose behavior cannot be predicted from either component alone. For instance, early adoption by a few agents can create localized pockets of reinforced positive attitudes that may or may not propagate outward, depending on network structure and threshold distribution.

We have added two new paragraphs:

- (1) At the end of the Related Models section, two paragraphs argue why bounded confidence, rather than DeGroot-style averaging (Golub & Jackson, 2010, 2012; Cheng et al., 2026) or voter model dynamics, is a theoretically necessary complement to threshold-based adoption, since only bounded confidence can sustain persistent opinion fragmentation as a long-run equilibrium property.
- (2) In the Emergent Properties of the Combined Model subsection (within the Simulation Model Specifications section), two new paragraphs identify two emergent structural properties that are unique to the BCAT framework: a dual-filter mechanism captured by the inequality  $0 \leq \text{GSI} \leq \text{FRI} \leq 1$ , and an adoption-to-opinion feedback loop in which adopted agents selectively reshape opinion trajectories among bounded-confidence-reachable neighbors (Algorithm 3, Scenarios 1.1 and 2.2).

In addition, we have added References to Golub & Jackson (2010, 2012), Cheng et al. (2026), and McPherson et al. (2001).

## R1-2. The Wedge Between Opinion and Adoption: Key Mechanisms

*The most interesting aspect of the paper is the divergence between opinion dynamics and adoption behavior. From the current model, I can identify at least two distinct channels. The first is a coordination failure channel: even when all agents hold positive attitudes, adoption requires a critical mass of neighbors to have already adopted, and this critical mass may fail to materialize if early adopters are too dispersed across the network. The second is an opinion clustering channel: bounded confidence can partition the population into opinion clusters, some of which settle below the positive-attitude threshold. It would be valuable for the authors to disentangle these channels more carefully.*

### **Response:**

Based on this suggestion, we designed three controlled mechanism decomposition experiments (MD-A, MD-B, MD-C) within the BCAT simulation framework, using parameter configurations that selectively disable one channel while preserving the other. We report results in a new subsection titled “Mechanism decomposition: Coordination failure versus opinion clustering,” and have added a new accompanying figure (Fig 10).

Details for the three controlled experiments include 30,000 total runs across regular lattice and small-world topologies. Three key findings are (1) the MD-A experiment reveals a phase transition in coordination failure between avg-of-thresholds = 30 (GSI = 0.976) and 40 (GSI = 0.026); (2) the MD-B experiment shows that opinion clustering alone reduced the FRI value to 0.600; and (3) the MD-C experiment demonstrates that the testimony effect recovers FRI to  $\approx 1.0$ , thus making coordination failure the dominant channel. A new Fig 10 presents the two-panel decomposition, and a new paragraph in the Discussion section describes the intervention implications of the two channels.

### R1-3. Empirical Grounding

*The paper's motivating examples – the Palm handheld, a Taiwanese film, and service-oriented architecture – are mentioned briefly in the introduction but not revisited. Given that the opinion–adoption wedge is the paper's central phenomenon, the empirical case for its relevance deserves fuller development.*

#### **Response:**

In response to this comment we have added case-specific interpretive paragraphs to the Discussion section following the parametric discussions most relevant to each case. Each paragraph maps the historically documented features of those cases onto BCAT model parameters and simulation outcomes. We do not claim formal calibration, but ground the theoretical interpretation in the known empirical characteristics of each example. No new simulations were required; these additions are interpretive elaborations based on existing sensitivity analysis results.

The Palm hand-held device discussion follows the avg-of-thresholds paragraph, with the mapping of coordination failure on regular lattice topology shown in Fig 5. The Taiwanese film example (The Bold, The Corrupt and The Beautiful) is discussed after the avg-of-attitudes paragraph, mapping the attitude baseline gap to the adoption plateau in Figs 8c-d. The SOA example is discussed after the bounded-confidence paragraph, mapping the professional vocabulary gap to bounded-confidence clustering failure in Figs 8a-b, and presenting the 2008 crisis as an example of exogenous avg-of-attitudes shock. A reference to Erl (2005) has been added for scholarly grounding of the SOA case.



#### R1-4. Alternative Theories and Distinguishing Predictions

*Other theories can also account for the opinion–adoption gap: rational herding and information cascades (Banerjee 1992), network externalities and coordination games (Katz and Shapiro 1985), and global games with heterogeneous thresholds (Morris and Shin 2003). Each mechanism implies different comparative statics. Discussing these distinctions, and suggesting how experiments or empirical tests might discriminate between them, would significantly enhance the paper’s contribution.*

#### **Response:**

In response to this comment we added six paragraphs to the Discussion section that systematically compares the BCAT model with the three alternative frameworks identified by the reviewer, and articulates the structural differences in mechanism, comparative statics, and empirical identifiability. We also discuss how future researchers can discriminate among these alternative frameworks, using time-series attitude data from sources such as social media sentiment analyses. Note that the BCAT model is the only one that generates detailed predictions about the temporal trajectory of attitude evolution prior to adoption outcomes.

The new paragraphs discuss several key distinguishing predictions. (1) Compared to Banerjee (1992), BCAT produces S-curve adoption (Figs 4, 6a) rather than step-function cascades, with social influences operating through transmission rather than Bayesian inference. (2) Compared to Katz & Shapiro (1985), BCAT predicts larger adoption gaps in clustered networks (see the Feature Importance data in Table 3), opposite of coordination game predictions. (3) Compared to Morris & Shin (2003), BCAT does not contain a higher-order belief mechanism – when  $FRI = 1.0$  and thresholds are low, adoption necessarily occurs (confirmed by MD-A experiments). References to Banerjee (1992), Katz & Shapiro (1985), and Morris & Shin (2003) have been added.

### **R1-Minor. Reference Error: Gong et al. (2020)**

*The Gong et al. (2020) reference contains what appears to be a corrupted title: "Hence, the source code used for this project may require significant modification to be applied to new studies."*

#### **Response:**

We confirmed that the title cited by this reviewer is a technical annotation from the BCAT simulation codebase that was inadvertently introduced into the title field during manuscript preparation. Thus, this is a field contamination error rather than an incorrect citation. The correct title is "A loud but harsh voice: The impact of celebrity endorsement on online WOM," published in *Journal of Management Science*, 2020, volume 33, issue 2, pages 114–126. The reference entry for Gong et al. (2020) has been corrected with the verified title and complete bibliographic information.

# Responses to Reviewer #2 comments

## R2-1. Discrepancy Between Deffuant Model and Simulation Implementation

*There is a fundamental discrepancy between the mathematical definitions and the simulation implementation. The authors cite the Deffuant et al. (2002) model, which is defined on a continuous opinion space with time-varying uncertainty. However, the BCAT implementation uses integer attitudes (1, 100) and a fixed bounded-confidence parameter. Methodologically, this makes the implementation a version of the HK (Hegselmann-Krause) model. The authors must reconcile the theoretical framing in Section 2 with the actual simulation.*

### Response:

We offer a two-part response to this technically precise and helpful observation.

Part 1: The integer attitude scale is a deliberate and mathematically equivalent discretization. Specifically, the integer scale [1, 100] is a uniform discretization of the continuous interval [0, 1] into 100 equally spaced levels, where each integer unit corresponds to an interval of 0.01. The bounded confidence parameter bounded-confidence follows the same convention: a value of 40 corresponds to a confidence threshold of  $\epsilon = 0.40$  in the standard literature. We made this design choice to facilitate sensitivity analysis, and to align the attitude scale with survey-style measurement instruments. We added a note in the Simulation Model Specifications section stating this equivalence.

Part 2: The opinion update mechanism corresponds to the simple bounded confidence Deffuant model, not the Relative Agreement variant. In our original manuscript, Section 2 introduces the Deffuant et al. (2002) Relative Agreement (RA) model, which features agent-specific, time-varying uncertainty parameters, and computes attitude adjustments as a function of confidence interval overlap (Equations 1 and 2). However, the BCAT opinion update procedure implements the rule  $\text{att\_new} = \text{round}(C + \mu \times (B - C))$ , where  $\mu$  is the fixed convergence-rate parameter. This rule does not involve any uncertainty parameter, does not compute confidence interval overlap, and does not update any agent-specific confidence bound. It is structurally equivalent to the simple Deffuant bounded confidence model (Deffuant et al., 2000) – that is, the pairwise random interaction variant in which two agents adjust their opinions by a fixed fraction  $\mu$  of their difference, provided the difference falls below a global fixed threshold  $\epsilon$ .

Though we retained the description of the RA model and its equations as context, we added a disambiguation sentence stating that the BCAT model does not implement the full RA update rule. A new paragraph describes the BCAT opinion update mechanism as a networked, discretized, adoption-status-conditioned extension of the simple Deffuant bounded confidence model (Deffuant et al., 2000). A

new comparative table (Table 1) summarizes the structural relationships among the HKmodel, the Deffuant RA model, and the proposed model across eight dimensions. We rewrote the Algorithm 3 description in the Simulation Model Specifications section to specify the fixed-rate convergence rule, the  $\varepsilon = \text{bounded-confidence}/100$  equivalence, and the absence of any agent-specific uncertainty parameter. Table 2 now includes a note on integer-to-continuous mapping. These clarifications affect the theoretical positioning of the model, but did not require any changes to the simulation itself.

## R2-2. Update Scheme: Synchronous vs. Asynchronous

*The manuscript leaves the update scheme ambiguous. Algorithm 3 suggests that at each tick, agents interact with one randomly chosen neighbor (asynchronous/sequential update). However, the overall simulation structure does not clarify if the adoption state is updated synchronously. Given that update order qualitatively affects convergence in opinion dynamics (cf. Berenbrink et al., 2024), the authors must explicitly define and justify the choice of synchronous vs. asynchronous updating.*

### **Response:**

BCAT uses a pseudo-concurrent update scheme based on the ask-concurrent command in NetLogo 4.0.5. According to ask-concurrent, all agents are processed in a randomly shuffled order within each tick, and each agent's state modifications take effect immediately. The Python 3 reimplementa-tion faithfully reproduces this behavior. This scheme lies between strictly synchronous updating (where all agents read old states before any writes occur) and strictly sequential updating with a fixed order (where position in the queue systematically advantages certain agents).

We chose this scheme because the random reshuffling of execution order at each tick ensures that no agent systematically benefits from its position in the processing sequence. Simulation randomness therefore comes from four well-defined stochastic elements: (1) the random selection of a neighbor for each agent per tick, (2) the initial random assignment of attitude and threshold values, (3) the random or clustered placement of pioneer agents, and (4) the random execution order within each tick. Because each source is controlled and reproducible given a fixed random seed, the sensitivity analysis results reported in the Results section reflect model parameter effects rather than systematic scheduling artifacts.

In response to this comment we added two new paragraphs to the Simulation Model Specifications section, immediately following the adoption decision-making description. The first explains the pseudo-concurrent update semantics of ask-concurrent (random-order sequential execution with immediate writes) and confirms that the Python 3 reimplementa-tion faithfully reproduces this behavior. The second states the methodological rationale: the random reshuffling of execution order during each tick prevents systematic scheduling bias, and identifies four controlled stochastic elements ensuring that the sensitivity analysis data reported in the Results section reflect model parameter effects.

### R2-3. Adoption Threshold Scaling (0–100 vs. 0–1)

*In Algorithm 3, the adoption threshold is treated as a percentage (0–100). In the standard diffusion literature (e.g., Watts 2004, Valente 1996), the threshold is strictly a fraction in (0–1). While not a fatal error, this non-standard scaling complicates comparisons with existing literature and makes the interpretation of Table 3 potentially misleading.*

#### **Response:**

As stated in our R2-1 response, the integer scale  $[1, 100]$  used throughout the BCAT model is a uniform 100-level discretization of the continuous interval  $[0, 1]$ , where each integer unit corresponds to an interval of 0.01. This scaling convention applies uniformly to both attitude values and adoption thresholds. An avg-of-thresholds value of 40, for instance, represents a mean adoption threshold of  $\theta = 0.40$  in the fractional notation standard in the diffusion literature (Watts, 2004; Valente, 1996), meaning that an agent adopts only when the proportion of its neighbors who have already adopted meets or exceeds 40%. The adoption decision rule of Algorithm 3 enforces this equivalence by computing the neighbor adoption proportion  $([0, 1])$  and comparing it against  $\theta/100$ . The two sides of the inequality therefore operate on identical scales, and the model's mathematical behavior is exactly equivalent to a formulation using fractional thresholds in  $[0.01, 1.00]$ .

We acknowledge that the original manuscript did not explicitly explain this scaling equivalence. We therefore added a clarifying statement in the Table 2 parameter description for avg-of-thresholds and std-of-thresholds, and appended a footnote to Table 3 providing the fractional equivalents for the threshold parameter values used in the sensitivity analysis. These changes do not require modifications to the simulation code or model behavior.

## R2-4. Incomplete Statistical Reporting in Table 3

*The reporting in Table 3 remains incomplete: (a) The specific feature importance algorithm is never identified. (b) Table 3 provides point estimates only; the authors must include standard errors or confidence intervals. (c) There appears to be a permutation error in the “All” row; values seem misaligned with individual network columns.*

### **Response:**

Regarding algorithm identification, we accept responsibility for this omission. We computed feature importance values using a Random Forest regression ensemble with 100 trees and a mean decrease in impurity (Gini importance) criterion, with all other hyperparameters at their default values. We have added this information to the Sensitivity Analysis subsection of our Results section.

Regarding uncertainty reporting, we revised Table 3 to include standard errors for the Random Forest feature importance values, computed as the standard deviation of per-tree importance values across the 100-tree ensemble divided by  $\sqrt{100}$ . All SE values are  $\leq 0.001$  (rounding to 0.00 at two decimal places), reflecting the high stability of importance estimates over large sample sizes ( $N = 15,876$  to  $100,548$ ). For the remaining four methods – multivariate regression, partial correlation, standardized regression, and parameter importance (Pearson correlation) – the reported values are deterministic analytical results given the input data, so standard errors are not applicable. We have added a footnote in Table 3 to clarify this distinction.

Regarding data alignment verification, we conducted a systematic audit of all values in Table 3, and discovered an error in the Parameter Importance section: values were inadvertently duplicated from the Standardized Regression section with rows and columns transposed. The Parameter Importance rows now report Pearson correlation coefficients between each parameter and adoption outcome variable, computed directly from the sensitivity analysis data. We also verified that the “All” row values across all five analysis methods are consistent with the aggregated dataset ( $N = 100,548$  observations across four network topologies). A footnote was added to Table 3 to explain why the corrected Parameter Importance values closely approximate the Standardized Regression coefficients – since the five input parameters vary independently in sensitivity analysis design, the predictors are approximately orthogonal, and the two measures mathematically equivalent under this condition.

## R2-5. Finite-Size Scaling Analysis

*The model runs with a fixed size of  $N=400$  agents, which is a very small sample, especially in the analysis of cascades and phase transitions. To ensure that the observed “Best Game No One Played” phenomenon is a robust collective behavior and not a stochastic artifact of a small system, the authors should provide a finite-size scaling analysis (e.g., testing  $N=1000, 5000$ ).*

### **Response:**

To establish robustness we conducted finite-size scaling experiments at  $N=400$  ( $20 \times 20$ ),  $N=900$  ( $30 \times 30$ ),  $N=1,600$  ( $40 \times 40$ ), and  $N=2,500$  ( $50 \times 50$ ), using identical parameter configurations (avg-of-attitudes=50, std-of-attitudes=15, bounded-confidence=50, avg-of-thresholds from 10 to 70). Our results confirm that the “best game no one played” phenomenon—high FRI coexisting with near-zero GSI—persisted at all system sizes, and that the observed adoption failure is a robust collective behavior rather than a stochastic artifact of the  $N=400$  baseline. The sharp phase transition at avg-of-thresholds between 30 and 40, and the dominance of avg-of-thresholds as the primary determinant of adoption outcomes, held across all system sizes. We added a note at the end of the sensitivity analysis subsection in our revised Results section. All experimental data are available in the public repository (DOI: [10.5281/zenodo.19216365](https://doi.org/10.5281/zenodo.19216365)).



## R2-6. Operational Definition of Critical Mass

*The authors frequently invoke “Critical Mass” (e.g., citing a value of 48 agents in Fig 6a) but fail to provide an operational definition within the model’s formal structure. It is unclear if these values are analytically derived or post hoc observations. The authors should define how “Critical Mass” is computed within the BCAT framework and whether it represents a first or second order phase transition in the adoption density.*

### **Response:**

In this revised manuscript, “critical mass” is used descriptively, based on Oliver et al. (1985), to refer to the minimum number of early adopters required to initiate a self-sustaining adoption cascade. The numerical values cited in Figs. 4 and 6a – 17 and 46 adopting agents, respectively – are informal observations from individual simulation runs, and are not recorded by the simulation as measured quantities. They serve as illustrative descriptions of the cascade dynamics observed in those runs.

“Critical point” ( $t^*$ ), which is directly recorded during simulations, has a precise operational definition within the BCAT framework: the first time step at which GSI exceeds 0.5 – that is, the first  $t$  at which adopters form a strict majority. Simulations compute and display this value in real time, and record it in each run’s output. When  $t^*$  is not achieved it is recorded as zero, indicating that the adoption process failed to reach majority diffusion – a “best game no one played” outcome. We added this explicit definition to our revised Results section.

The above-described mechanism decomposition experiments (see response to R1-2) document a sharp transition in GSI between avg-of-thresholds = 30 and 40, and the finite-size scaling experiments (see response to R2-5) confirm that this transition persists across system sizes  $N = 400$  through  $N = 2,500$ .

## R2-7. Real-World Cases and Comparison with Alternative Coupled Models

*The real-world cases (Palm, Taiwanese cinema) are used only as anecdotal motivation. The authors should explicitly discuss the limitations of mapping the 1–100 attitude scale to real-world consumer data. Furthermore, they should briefly compare BCAT’s quantitative behavior with existing coupled models (e.g., voter models with inertia or modified SIS/SIR frameworks).*

### Response:

For the first part of this comment, please see our response to the third comment of reviewer #1, and the three case-specific paragraphs we added to the Discussion section concerning motivating cases with qualitative parameter mappings.

Regarding the discretization equivalence of the 1–100 attitude scale, please see our responses to comments 1 and 3 to reviewer #2 above.

Regarding the comparison with alternative coupled models, we added a paragraph to our Discussion section (following the alternative theories comparison in response to R1-4) that contrasts BCAT with voter models with inertia and SIS/SIR epidemic frameworks. We note that voter models with inertia (Stark et al., 2008) generate metastable heterogeneous states but operate on binary opinion spaces, thus precluding the continuous attitude variation that allows BCAT to capture nuanced scenarios in which agents hold positive but insufficiently strong attitudes. SIS/SIR epidemic frameworks treat adoption as a probabilistic contagion process without an independent attitude layer; any contact with an adopter may trigger adoption regardless of private sentiment. In contrast, BCAT’s two-layer architecture—requiring both positive attitude and sufficient neighbor adoption—enables the emergence of the opinion–adoption gap, a phenomenon structurally absent from single-layer contagion models.

## R2-8. Reference Error: Gong et al. (2020)

*Reference “Gong et al. (2020)” has a mistake. The title field contains a fragment of a footnote regarding NetLogo source code modification. Correct or replace this bibliographic entry.*

### Response:

Please see our “Minor” response to referee #1 above.

## References cited in this response letter

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